

Incorporating LLMs in our AI literacy curriculum to meet youth's critical needs

Our project *planet+AI* was funded by NSF AISL DRL-2314155. We conducted our first-year summer program in 2024 and reached out to 47 local youth. We plan to reach out to another 100 youth in the next two years. Now we request reallocation of part of the rest of the participant support funds (we originally proposed to reach out 360 youth in three years) to adding a new co-PI with expertise in LLMs to our Project Team so we will incorporate large language models (LLMs) in our AI literacy curriculum. LLMs has been expected and desired by our youth since the beginning of our project. We request that we add Dr. Meng Jiang, an Associate Professor in the Department of Computer Science and Engineering at the University of Notre Dame (ND), to our Project Team. The reallocation will enable us to provide our youth crucial and timely experiences and perspectives of learning and using LLMs for *planet+AI*, and meet youth's critical needs. The reallocation is necessary and reasonable, and does not change in objectives or scope of our proposed work. The reallocation will allow us to more effectively utilize funds and resources, and will enable us to reach out even more youth in informal and formal learning settings.

The observations and feedback from our public webinar series [1], public lecture series [2], and our summer program last year [3] shows that youth are very keen to learn more about large language models (LLMs). However, our proposed AI literacy curriculum lacks the LLMs component due to two reasons: (1) Our Project Team proposed *planet+AI* prior to the end of 2022 when ChatGPT (GPT-3.5) was launched [4] and thus our current AI literacy curriculum has been focusing on computer vision and Convolutional Neural Networks (CNNs). (2) Our current Project Team does not have an expert in the fields of LLMs and clearly, it's crucial to recruit an expert in LLMs in our Project Team.

Since our *planet+AI* started, PI Wang has been proactively reaching out to faculty members in the Department of Electrical Engineering and Computer Science (EECS) at University of Tennessee, Knoxville (UTK), including Dr. Jim Plank, Dr. Hairong Qi, Dr. Weizi Li, Dr. Jian Liu and Dr. Fei Liu. However, we have not been able to find an LLM expert at UTK to support our youth, which results in our youth not being able to seek and receive sufficient support timely when encountering problems related to LLMs. For example, participating youth in our summer program raised many questions in LLMs, such as *why and how Copilot on Google Colab is able to write code*, and our originally proposed curriculum did not cover anything related to LLMs. Two youth teams Wang has mentored during the previous year have been struggling badly due to lack of LLM expertise in our Project Team. The DAISY team (see supplementary materials) built an AI chatbot last year on natural disaster responses were not able to fine-tune models due to lack of knowledge and resources about LLM training; the WATER team (see supplementary materials) would benefit greatly with an LLM expert on how to use new LLMs techniques to discover anomalies in the water dataset. A few faculty members and PhD students at UIUC helped the DAISY team and the WATER team, including Patrick Jiang and Revanth Gangi Reddy; however, they would have done so much better if we could have an LLM expert on our Project Team. The WATER team's work has been widely welcomed by many local youth due to its hands-on nature- youth conducted field work, lab work, and data reduction themselves, which deepens their understanding in earth and planetary data, tabular data, geospatial data, high-dimensional data, and PCA algorithm. It's easy to notice that the WATER team took great efforts in exploring new ways, particularly LLMs, in their work.

Dr. Meng Jiang has been dedicating his time and supporting our program and our youth with his expertise in LLMs [5, 6, 7, 8, 9, 10] since he knew of our difficulty. Jiang's LLMs course for graduate students has been widely welcomed with a rating of 5/5 at ND, covering topics, including LLMs concepts (e.g., GPT, CoT, in-context learning, instruction fine-tuning, RAG, multi-agent, halluci-

nation), various tasks and datasets related to language models, LLMs architectures, LLMs training techniques, reasoning methods, LLM applications (e.g., education, recommender systems), and challenges in LLMs for social good. Jiang has been customizing his course materials for our youth and has been sharing his course materials with us. He has been spending his time with our Project Team in group meetings on how to use AI/ML, particularly LLMs, with earth and planetary data and images. He has been collaborating with Project Team in improving our current AI literacy curriculum, particularly using LLMs techniques with water dataset, chondrite dataset, and other planetary datasets. He has been answering our youth's questions in Discord Channel and via Email. Jiang has been liked and wanted by our youth. With a dedicated co-PI, Dr. Jiang, on our Project Team, we will be able to update our AI literacy curriculum to meet the critical needs for learning, understanding and using LLMs.

Adding Dr. Jiang to our Project Team as a co-PI and reallocation part of the funds will enable our youth and educators in schools and in informal learning settings to have access to AI platforms that support LLMs deployment, ensuring youth and educators to have seamless interaction with AI, particularly LLMs tools which has been desired by our youth since the beginning of our project. The APIs of text and image generation, such as GPT Vision API [11] on the OpenAI platform, are important resources to designing and implementing lesson plans that incorporate LLM-based activities, aligning them with planetary science components. We will also use Google Colab to teach students to use open-weight vision-language models such as LLaVA-1.5 [12], QWen2.5-VL [13], and InternVL2.5 [14].

Adding Dr. Jiang to our Project Team as a co-PI will make possible that we reach out to many informal learning settings, including museums and out-of-school organizations, including the American Museum of Science and Energy in Oak Ridge, TN, and Harvard Mineralogical and Geological Museum as many informal learning institutions have been keen to incorporate emerging AI, particularly LLMs, in their outreach programs. In turn, our Project Team with expertise in earth and planetary sciences, computer vision and LLMs, will be reaching out to thousands of youth, educators, and public audiences, even more than originally proposed, through close collaboration to meet what is crucially needed by the public at this time.

With Dr. Jiang on our Project Team as a co-PI, we will develop and conduct an innovative LLMs-enhanced survey method to conduct our research and large-scale educational data analysis, particularly in informal STEM learning, together with Dr. Yanran Chen, a postdoctoral research associate at ND, who has been supporting our linear algebra. With Jiang's expertise in LLMs and Chen's expertise in educational psychology and designing and conducting AI-enhanced large-scale survey [15, 16, 17, 18, 19, 20], we will incorporate LLMs as essential tools for brainstorming, refining, and expanding our question sets. By leveraging the natural language generation and editing capabilities of LLMs, we will quickly and systematically develop questions that tap into a broader range of youth experiences, knowledge levels, and learning styles than would be possible through conventional methods alone. With Jiang and Chen on our Project Team, we will craft a concise set of fundamental, structured questions that cover key performance indicators and relevant feedback dimensions. These questions will address critical areas such as youth's understanding of curriculum content, their engagement with learning materials, and their overall satisfaction with instructional methods. We will feed the fundamental questions into an LLM, which will then generate a larger, more diverse pool of survey items designed to accommodate variations in learning backgrounds and educational contexts. This expanded question bank will ensure that we capture a range of perspectives, from youth who may have extensive prior knowledge in the subject area to those who are relatively new. After each youth submits their coursework, we will analyze the content of their submission to identify unique learning patterns, topic interests, and potential knowledge gaps. The LLM, using this information, will then select the most targeted questions from our expanded bank, ensuring that each survey is tailored to youth's specific background,

performance, and skill set. This level of granularity not only encourages more thoughtful and engaged responses but also fosters a deeper understanding of how students are processing and applying the material. By incorporating LLMs tools adn with Jiang and Chen’s expertise, we will yield a uniquely robust, data-rich picture of student learning, enabling us to refine and adapt educational strategies with unprecedented precision.

With Jiang as a co-PI on our Project Team, we will update our current curriculum, integrating LLMs, computer vision, with earth and planetary data and images. Our proposed **Planet+AI: Data, Algorithms, and Compute** will include:

- **Unit 1: Planetary Data and Planetary Images, High-dimensional Data, Big Data:** Under our guidance, participating youth conduct a hand-on ICP-MS for water project which will give youth a deeper understanding geochemical data.
- **Unit 2: PCA Algorithm with High-dimensional Data:** Youth learn the PCA algorithm through basic linear algebra and statistics, and use PCA with water data.
- **Unit 3: Supervised Learning and Unsupervised Learning for Chondrite Images and Data:** Youth learn CNNs, PCA, clustering, classification and chondrules segmentation.
- **Unit 4: 3D Reconstruction and SfM Photogrammetry:** Youth shoot photos for SfM photogrammetry and use Reality Capture with GPUs to create 3D models of rocks, minerals, and meteorites, including porphyry which was brought back from a recent field trip to Norway. Youth use same method to create 3D outcrop models with drone and SfM-based geological mapping. Youth learn and use the same technique to create 3D models with planetary images sent back from NASA’s previous mission.
- **Unit 5: Basics of LLMs: Tokenization and Autoregression:** Students use tokenization techniques to break down sample texts. Students will train GPT-2 models by predicting the next token in a Geoscience corpus and analyze the LLM behaviors.
- **Unit 6: Accurate Extraction of Tabular Data from Research Papers:** Students learn to use information extraction techniques such as Camelot, PdfPlumber, and olmOCR. They will also learn to use LLMs (e.g., ChatGPT, Llama, Mistral) to extract tabular data from PDF files.
- **Unit 7: Chatbot and Retrieval-Augmented Generation (RAG):** Students will build a chatbot with LLMs (e.g., Llama, Mistral) to answer questions on earth and planetary topics. When the responses are limited to knowledge, with the corpus and tabular data, students will develop a RAG framework to enhance the chatbot knowledge and performance.
- **Unit 8: Interpretable Anomaly Detection on Geochemical Tabular Data:** Students will use outlier detection techniques (e.g., DBSCAN [21], ECOD [22]) to find geochemical anomalies. They will use LLMs to explain the anomalies in natural language.

Jiang and his GRA, through close collaboration with other members on our Project Team, will design, develop, and teach units 5-8.

References

- [1] planet+AI. Planet+ai public webinar series, 2023. <https://planet-ai-2023.github.io/2023/09/11/webinar-sep-24-interstellar/> [Accessed: (March 11, 2025)].
- [2] planet+AI. Planet+ai public lecture series, 2025. <https://planet-ai-2025.github.io/2025/01/02/linalg/> [Accessed: (March 11, 2025)].
- [3] planet+AI. Planet+ai summer program 2024, 2024. <https://planet-ai-2023.github.io/2024/06/01/summer-2024/> [Accessed: (March 11, 2025)].
- [4] OpenAI. Introducing chatgpt, 2022. <https://openai.com/index/chatgpt/> [Accessed: (March 11, 2025)].

- [5] Tianwen Jiang, Tong Zhao, Bing Qin, Ting Liu, Nitesh V Chawla, and Meng Jiang. The role of "condition" a novel scientific knowledge graph representation and construction model. In *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pages 1634–1642, 2019.
- [6] Daheng Wang, Prashant Shiralkar, Colin Lockard, Binxuan Huang, Xin Luna Dong, and Meng Jiang. Tcn: Table convolutional network for web table interpretation. In *Proceedings of the web conference 2021*, pages 4020–4032, 2021.
- [7] Wenhao Yu, Dan Iter, Shuohang Wang, Yichong Xu, Mingxuan Ju, Soumya Sanyal, Chenguang Zhu, Michael Zeng, and Meng Jiang. Generate rather than retrieve: Large language models are strong context generators. In *International Conference on Learning Representations*, 2023.
- [8] Noah Ziemis, Wenhao Yu, Zhihan Zhang, and Meng Jiang. Large language models are built-in autoregressive search engines. In *The Annual Meeting of the Association for Computational Linguistics*, 2023.
- [9] Zheyuan Liu, Guangyao Dou, Zhaoxuan Tan, Yijun Tian, and Meng Jiang. Towards safer large language models through machine unlearning. In *The Annual Meeting of the Association for Computational Linguistics*, 2024.
- [10] Zheyuan Liu, Guangyao Dou, Zhaoxuan Tan, Yijun Tian, and Meng Jiang. Machine unlearning in generative ai: A survey. *arXiv preprint arXiv:2407.20516*, 2024.
- [11] Wenhao Wu, Huanjin Yao, Mengxi Zhang, Yuxin Song, Wanli Ouyang, and Jingdong Wang. Gpt4vis: what can gpt-4 do for zero-shot visual recognition? *arXiv preprint arXiv:2311.15732*, 2023.
- [12] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *NeurIPS*, 2023.
- [13] Jinze Bai, Shuai Bai, Shusheng Yang, Shijie Wang, Sinan Tan, Peng Wang, Junyang Lin, Chang Zhou, and Jingren Zhou. Qwen-vl: A versatile vision-language model for understanding, localization, text reading, and beyond. *arXiv preprint arXiv:2308.12966*, 2023.
- [14] Zhe Chen, Jiannan Wu, Wenhao Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qionglong Zhang, Xizhou Zhu, Lewei Lu, et al. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24185–24198, 2024.
- [15] Joseph Check and Russell K Schutt. Survey research. *Research methods in education*, 26:159–185, 2012.
- [16] Anthony R Artino Jr, Jeffrey S La Rochelle, Kent J Dezee, and Hunter Gehlbach. Developing questionnaires for educational research: Amee guide no. 87. *Medical teacher*, 36(6):463–474, 2014.
- [17] Jonathan C Hilpert and Gwen C Marchand. Complex systems research in educational psychology: Aligning theory and method. *Educational psychologist*, 53(3):185–202, 2018.
- [18] Mark Prendergast and Niamh O’Meara. Questionnaire design and implementation. In *Perspectives in Contemporary STEM Education Research*, pages 120–128. Routledge, 2022.
- [19] René F Kizilcec. To advance ai use in education, focus on understanding educators. *International Journal of Artificial Intelligence in Education*, 34(1):12–19, 2024.
- [20] Yoon-Sung Choi. Earth science simulations with generative artificial intelligence (genai). *Journal of University Teaching and Learning Practice*, 22(1):1–24, 2025.

- [21] Martin Ester, Hans-Peter Kriegel, Jörg Sander, Xiaowei Xu, et al. A density-based algorithm for discovering clusters in large spatial databases with noise. In *kdd*, volume 96, pages 226–231, 1996.
- [22] Zheng Li, Yue Zhao, Xiyang Hu, Nicola Botta, Cezar Ionescu, and George H Chen. Ecod: Un-supervised outlier detection using empirical cumulative distribution functions. *IEEE Transactions on Knowledge and Data Engineering*, 35(12):12181–12193, 2022.



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RESEARCH

Administration

PI
Sponsor
Project Start Date
Project End Date
Total Sponsor Obligated

Meng Jiang
UTennessee
5/1/2025
8/31/2026
\$276,238

Grant
Fund
Organization
Program
F&A
0
0
0
0
56.5%

	Sponsor Year 1 M1-4	Sponsor Year 2 M5-16	Sponsor Year 3	Sponsor Year 4	Sponsor Year 5	Sponsor Total
Salaries and Wages						
Senior Personnel AY/CY Salary	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Senior Personnel Summer Salary	\$ 16,706	\$ 17,374	\$ -	\$ -	\$ -	\$ 34,080
Post Doctoral Associates	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Graduate Students	\$ 13,867	\$ 43,264	\$ -	\$ -	\$ -	\$ 57,131
Undergraduate Students	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Other Personnel	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Fringe Benefits	\$ 6,032	\$ 10,167	\$ -	\$ -	\$ -	\$ 16,199
Equipment	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Travel						
Domestic	\$ -	\$ 3,000	\$ -	\$ -	\$ -	\$ 3,000
Foreign	\$ -	\$ 4,000	\$ -	\$ -	\$ -	\$ 4,000
Participant/Trainee Support Costs	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Other Direct Costs						
Materials and Supplies	\$ 20,217	\$ 20,217	\$ -	\$ -	\$ -	\$ 40,434
Publication Costs	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Consultant Services	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
ADP/Computer Services	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Subawards/Consortium/Contractual Costs	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Facility or Equipment Rental/User Fees	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Alterations and Renovations	\$ -	\$ -	\$ -	\$ -	\$ -	\$ -
Other Direct	\$ 10,833	\$ 10,833	\$ -	\$ -	\$ -	\$ 21,666
Total Direct Costs	\$ 67,655	\$ 108,855	\$ -	\$ -	\$ -	\$ 176,510
MTDC Base	\$ 67,655	\$ 108,855	\$ -	\$ -	\$ -	\$ 176,510
Indirect Costs	\$ 38,225	\$ 61,503	\$ -	\$ -	\$ -	\$ 99,728
Total Direct and Indirect Costs	\$ 105,880	\$ 170,358	\$ -	\$ -	\$ -	\$ 276,238

BUDGET JUSTIFICATION

University of Notre Dame

A fiscal year at the University of Notre Dame is defined as the time period from July 1st through June 30th annually. For the purposes of this proposal, one project year is defined as 365 days from the proposed start date of May 1, 2025 and proposed end date of August 31, 2026.

Personnel

Dr. Meng Jiang, PI (1.0 summer month)

Dr. Jiang, Associate Professor of Computer Science and Engineering, will serve as the PD/PI at the University of Notre Dame for the proposed project and will be responsible for all aspects of its development, implementation, and management.

TBN, Graduate Student (4 calendar months year 1, 12 calendar months year 2)

Salary support is requested for a full-time Graduate Research Assistant from the Department of Computer Science and Engineering to be responsible for enhancing AI curriculum with concepts and methods of large language models for high school youth to explore planetary images.

Salaries and wages (including stipends) are based on the University of Notre Dame's budget for fiscal year 2025 with 4 percent (4%) annual increment compensation.

Fringe Benefits

University of Notre Dame fringe benefits are calculated as a percentage of requested salary dollars using the appropriate rate for each personnel category. The rates have been approved by our federal cognizant agency, the Department of Health and Human Services (DHHS), and are documented in the university's negotiated rate agreement.

<i>Description</i>	<i>FY2025 Fringe Rates</i>
<i>Regular Full-Time Faculty</i>	24.9%
<i>Graduate Students</i>	13.5%

Equipment – Not Applicable

Travel – \$3,000 domestic in year 2; \$4,000 international in year 2

\$3,000 total is requested for the PI and graduate student to present research findings at domestic professional meetings. The major domestic conference is:

- ASEE Annual Conference & Exposition (American Society for Engineering Education): Charlotte, North Carolina, June 21 - 24, 2026.

\$4,000 total is requested for the PI and graduate student to present research findings at international professional meetings. Major domestic conferences are:

- Learning Analytics and Knowledge (LAK) Conference: Bergen, Norway, April 27 - May 1, 2026.

All costs include airfare, lodging, meals and ground transportation based on prior experience, University of Notre Dame Travel Policy, and per diem established by gsa.gov and/or U.S. Department of State.

Participant Support – Not Applicable

Materials & Supplies – \$20,217 year 1, \$20,217 year 2

\$40,434 total is requested for consumable research supplies including (a) the cost of Google Colab supported by Google Cloud GPUs, (b) the cost of Vision APIs (text, audio, and image generation) from OpenAI Platform.

Cost of Google Colab: With Google Colab Pro+, 500 compute units can support about 38.5 hours of A100 GPU (40GB) usage, which cost \$49.99 (see “Colab Pricing.pdf”). That is \$1.29 per GPU hour, which is the same as Lambda On-Demand Cloud (see “8xNVIDIA A100 SXM 40GB” in “Lambda Pricing.pdf”). So we use Google Colab. Each training curriculum offering takes 3 weeks, which consumes **\$14,023** (30 students, 1 GPU card each student, 360 hours).

Cost of Vision APIs: The training curriculum provides each student 1,000 images to do the study. Each image consumes about 765 tokens. Each curriculum is offered to 30 students. That’s about **22.95M** input tokens. The outputs of each image would have around 500 words (650 tokens). On average, one will attempt at least 5 kinds of prompts to tune the outputs. That’s about **97.5M** output tokens. Based on the pricing in March 2025 (see “OpenAI Pricing.pdf”), the monthly cost is: $\$15/1M * 22.95M + \$60/1M * 97.5M = \mathbf{\$6,194}$.

Publications – Not Applicable

Other Direct Costs: Human Subject Payments – \$10,833 year 1, \$10,833 year 2

\$21,666 is requested to support incentive payments to human subject study participants. The funds will compensate the human subject participant for their time and expertise in survey methodology, data collection, and data analysis. The budget is based on an estimated effort of approximately 300 human hours per year, considering competitive compensation, expertise, and the complexity of the tasks involved.

The participants, with rich experience in survey design and data collection, are responsible for designing and developing surveys to collect data from high school youth participating in the AI for planetary science curriculum. The participant role is critical in ensuring the development of well-structured and scientifically valid survey instruments to assess students' engagement, learning outcomes, and perceptions of AI applications in planetary science.

Indirect Costs

Indirect costs are charged as a percentage of the proposed project’s modified total direct costs (MTDC). MTDC excludes equipment, capital expenditures, charges for patient care, rental costs, tuition remission, scholarships and fellowships, participant support costs, and the portion of each subaward in excess of \$25,000. The University’s current indirect cost rate for on campus research activities is 56.5%. This rate was established with the Department of Health and Human Services and became effective on April 17, 2024.

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Ability to use a terminal with the connected VM.

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Purchase more as you need them.
- ✓ Faster GPUs
Priority access to upgrade to more powerful premium GPUs.
- ✓ Background execution
With compute units, your actively running notebook will continue running for up to 24hrs, even if you close your browser.

GPUs	VRAM/GPU	vCPUs	RAM	STORAGE	PRICE/GPU/HR*
On-demand 8x NVIDIA H100 SXM	80 GB	208	1800 GiB	22 TiB SSD	\$2.99 / GPU / hr
Reserved 8x NVIDIA H100 SXM	80 GB	208	1800 GiB	22 TiB SSD	CONTACT SALES
On-demand 8x NVIDIA A100 SXM	80 GB	240	1800 GiB	19.5 TiB SSD	\$1.79 / GPU / hr
On-demand 8x NVIDIA A100 SXM	40 GB	124	1800 GiB	5.8 TiB SSD	\$1.29 / GPU / hr
On-demand 8x NVIDIA Tesla V100	16 GB	88	448 GiB	5.8 TiB SSD	\$0.55 / GPU / hr

*plus applicable sales tax



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OpenAI o1

Frontier reasoning model that supports tools, Structured Outputs, and vision | 200k context length

Price

Input:
\$15.00 / 1M tokens

Cached input:
\$7.50 / 1M tokens

Output:
\$60.00 / 1M tokens

OpenAI o3-mini

Small cost-efficient reasoning model that's optimized for coding, math, and science, and supports tools and Structured Outputs | 200k context length

Price

Input:
\$1.10 / 1M tokens

Cached input:
\$0.55 / 1M tokens

Output:
\$4.40 / 1M tokens

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Calculating costs

Image inputs are metered and charged in tokens, just as text inputs are. The token cost of an image is determined by two factors: size and detail.

Any image with `"detail": "low"` costs 85 tokens. To calculate the cost of an image with `"detail": "high"`, we do the following:

- Scale to fit in a 2048px x 2048px square, maintaining original aspect ratio
- Scale so that the image's shortest side is 768px long
- Count the number of 512px squares in the image—each square costs **170 tokens**
- Add **85 tokens** to the total

Cost calculation examples

- A 1024 x 1024 square image in `"detail": "high"` mode costs 765 tokens
 - 1024 is less than 2048, so there is no initial resize.
 - The shortest side is 1024, so we scale the image down to 768 x 768.
 - 4 512px square tiles are needed to represent the image, so the final token cost is $170 * 4 + 85 = 765$.
- A 2048 x 4096 image in `"detail": "high"` mode costs 1105 tokens
 - We scale down the image to 1024 x 2048 to fit within the 2048 square.
 - The shortest side is 1024, so we further scale down to 768 x 1536.
 - 6 512px tiles are needed, so the final token cost is $170 * 6 + 85 = 1105$.
- A 4096 x 8192 image in `"detail": "low"` most costs 85 tokens
 - Regardless of input size, low detail images are a fixed cost.

We process images at the token level, so each image we process counts towards your tokens per minute (TPM) limit. See the calculating costs section for details on the formula used to determine token count per image.

Chat Completions

[Image input](#)[Image input requirements](#)[Specify input detail](#)[Multiple image inputs](#)[Limitations](#)[Calculating costs](#)